

Trading Strategy Proposals for 5-Minute Crypto Resolution Markets on Polymarket

Classification: Implementation Blueprint
Date: May 2026
Scope: Binary / multi-outcome contracts resolving on crypto price thresholds within 5-minute windows

0. Preliminaries: Why 5-Minute Crypto Markets Are Structurally Different

A 5-minute crypto Polymarket contract is not a long-horizon prediction market. It is a **micro-duration derivative** whose payoff depends on whether a spot price S_T crosses a strike K at resolution time T , where $T - t_0 = 5$ minutes. This has profound implications:

- The "true probability" is approximately computable.** Unlike political markets, the underlying (BTC, ETH spot price) is observable in real-time with sub-second latency from CEX order books. You can compute $P(S_T > K \mid S_t, \sigma_t, \tau)$ using a diffusion approximation.
- Latency arbitrage is the dominant edge source.** The Polymarket order book updates slower than Binance/Coinbase spot. A price move on Binance at t may not be reflected on Polymarket until $t + \delta$ where $\delta \in [2, 30]$ seconds.
- Liquidity is thin and stale.** Market makers on 5-min contracts often quote wide spreads and update infrequently. This creates both opportunity (stale quotes) and risk (adverse selection, slippage).
- The market is effectively a discretized binary option.** Pricing theory from options markets applies, but with bounded $[0,1]$ probability space and discrete resolution.

Notation

Symbol	Meaning
S_t	Spot crypto price at time t (from CEX)
K	Strike/threshold of the Polymarket contract
p_t	Polymarket YES share price (implied probability) at time t
τ	Time to resolution (seconds remaining)
σ_t	Realized volatility estimate over recent window
$\Delta_t = S_t - K$	Moneyness in spot-price space
h	Trading horizon (seconds before resolution at which you enter)

Strategy 1: Spot-Led Latency Arbitrage (SLA)

1.1 Classification

Type: Heuristic rule-based system

Edge source: Information latency between CEX spot feeds and Polymarket order book

Complexity: Low-Medium

Capital efficiency: High (short holding periods, high turnover)

1.2 Rationale

The core insight is that Polymarket 5-minute markets are **price-takers** relative to the CEX spot market. When BTC moves 0.3% on Binance in 10 seconds, the fair probability of "BTC > \$K\$ at resolution" changes materially, but Polymarket market makers may lag by δ seconds. During this window, the Polymarket price is stale and exploitable.

Why this works structurally:

- CEX crypto markets have far more liquidity and faster participants than Polymarket
- Polymarket MM algorithms must balance inventory risk across many markets and cannot update as fast as a dedicated CEX HFT system
- The 5-minute resolution window means even a small spot move can shift the "true" probability by 5-20 percentage points

1.3 Signal Construction

At each decision time t , compute the **fair probability** $\hat{\pi}_t$ using a Black-Scholes-style binary call approximation:

$$\hat{\pi}_t = \Phi\left(\frac{\ln(S_t / K) + (\mu - \sigma_t^2/2)\tau}{\sigma_t \sqrt{\tau}}\right)$$

where:

- Φ is the standard normal CDF
- μ is the short-term drift (can be set to 0 for a pure diffusion assumption over 5 minutes)
- σ_t is estimated from a rolling window of 1-minute returns on the CEX
- τ is seconds remaining to resolution

The **mispricing signal** is:

$$\epsilon_t = \hat{\pi}_t - p_t$$

Decision rule:

- If $\epsilon_t > \theta_{entry}$ AND p_t^{ask} is available at a price $< \hat{\pi}_t - c$ (where c is estimated cost), BUY YES at ask.
- If $\epsilon_t < -\theta_{entry}$ AND p_t^{bid} is available at a price $> \hat{\pi}_t + c$, SELL YES (or BUY NO) at bid.

1.4 Empirically Discoverable Parameters

Parameter	Symbol	Discovery Method	Expected Range
Entry threshold	θ_{entry}	Grid search on historical data, optimizing for Sharpe ratio net of costs	0.03 - 0.10
Volatility window	w_{σ}	Rolling window length for σ_t estimation; test 30s, 60s, 120s, 300s	30 - 300 seconds
Drift parameter	μ	Can be set to 0 or estimated from EMA of 1-min returns; test sensitivity	0 or small
Cost estimate	c	Empirical spread + fee estimate; measure from historical order book data	0.01 - 0.04
Minimum time to resolution	τ_{min}	Don't trade if $\tau < \tau_{\text{min}}$ (too illiquid); test 30s, 60s, 120s	30 - 120 seconds
Maximum position size	q_{max}	Risk management; test impact on PnL	50 - 500 shares

1.5 Concrete Example

Setup: BTC spot = \$68,450. Polymarket contract: "Will BTC be above \$68,500 at 14:05 UTC?" (resolution in 3 minutes).

1. Current spot moves to \$68,520 on Binance (a +\$70 move in 15 seconds).
2. Compute $\hat{\pi}_t$: $\sigma_t = 0.001$ (per minute), $\tau = 180$ s. $\ln(68520/68500) = 0.000292$.
 $\hat{\pi}_t = \Phi(0.000292 / (0.001 \times \sqrt{3})) \approx \Phi(0.169) \approx 0.567$
3. Polymarket still shows YES at bid=0.42, ask=0.46 (implied prob ~0.44).
4. Signal: $\epsilon_t = 0.567 - 0.44 = 0.127$. This exceeds $\theta_{\text{entry}} = 0.05$.
5. **Action:** BUY YES at ask = \$0.46. Expected value per share = $0.567 - 0.46 - 0.02 = +0.087$.
6. Hold to resolution. If BTC stays above \$68,500, share resolves to \$1.00. Profit = \$0.54 per share.

1.6 Statistical Backing

The strategy relies on the **efficient market hypothesis violation due to latency**. Academic work on cross-venue latency arbitrage (e.g., Budish, Cramton & Shim 2015; Baron et al. 2019 on HFT) shows that price discrepancies between faster and slower venues persist for short windows. The key statistic to validate is the **autocorrelation of the mispricing signal** ϵ_t : if it decays within δ seconds, the window exists. Measure the half-life of ϵ_t autocorrelation; if $t_{1/2} > 3$ seconds, the strategy is potentially viable.

1.7 Failure Modes and Mitigations

Risk	Mitigation
Spot reversal before resolution	Only enter when τ is small relative to spot distance from strike
Adverse selection (MM pulls liquidity)	Use limit orders with stale-quote detection
Polymarket API latency	Co-locate infrastructure; use WebSocket feeds
Volatility regime change	Adaptive σ_t estimation with exponential weighting

Strategy 2: Probability Mean-Reversion After Overreaction (PMR)

2.1 Classification

Type: Statistical model (time-series + distributional)
Edge source: Behavioral overreaction by retail traders in thin prediction markets
Complexity: Medium
Capital efficiency: Medium (requires holding positions for 1-4 minutes)

2.2 Rationale

Retail participants on Polymarket tend to **overreact to salient information**. When a large spot candle appears (e.g., BTC drops 0.5% in 30 seconds), traders rush to update their probability estimates, often overshooting the fair value. This creates a short-term mean-reversion opportunity.

Why this works:

- Prediction markets with retail-heavy participation exhibit documented overreaction (Page 2012; Camerer et al. on prospect theory in markets)
- The 5-minute window means traders are reacting emotionally to noise, not fundamentals
- The "true" probability can be continuously computed from the spot price, making overreaction measurable in real-time
- Mean-reversion in probability space is analogous to short-term reversal in equity markets (Lehmann 1990; Lo & MacKinlay 1990)

2.3 Statistical Model

Define the **overreaction indicator** as the rate of change of the implied probability:

$$\dot{p}_t = \frac{p_t - p_{t-w}}{w}$$

where w is a lookback window (e.g., 15 seconds).

Define the **fair probability rate of change**:

$$\dot{\hat{\pi}}_t = \frac{\hat{\pi}_t - \hat{\pi}_{t-w}}{w}$$

The **overreaction magnitude** is:

$$O_t = p_t - \dot{\hat{\pi}}_t$$

When O_t exceeds a threshold (in standard deviations, estimated from a rolling window), we expect mean-reversion.

Signal: $S_t = -\text{sign}(O_t) \cdot f(|O_t|)$

where f is a monotonically increasing function (e.g., clipped linear or sigmoid).

Model for reversion dynamics: Fit an Ornstein-Uhlenbeck (OU) process to the mispricing $\epsilon_t = p_t - \hat{\pi}_t$:

$$d\epsilon_t = -\kappa \epsilon_t dt + \nu dW_t$$

The mean-reversion speed κ and noise ν are estimated from historical data. The expected reversion over horizon h is:

$$E[\epsilon_{t+h} | \epsilon_t] = \epsilon_t e^{-\kappa h}$$

Trade when $E[\text{reversion profit}] > \text{cost}$:

$$|\epsilon_t| (1 - e^{-\kappa h}) > c$$

2.4 Empirically Discoverable Parameters

Parameter	Symbol	Discovery Method	Expected Range
Lookback window for rate of change	w	Test 10s, 15s, 30s, 60s; optimize for signal IC (information coefficient)	10 - 60 seconds
OU reversion speed	κ	MLE estimation on ϵ_t time series per market	0.1 - 2.0 per minute
OU noise	ν	Joint MLE with κ	Data-dependent
Overreaction threshold (z-score)	z_{θ}	Grid search on standardized O_t ; optimize Sharpe	1.5 - 3.0
Minimum τ to enter	$\tau_{\min}$	Same as Strategy 1	60 - 180 seconds
Position holding time	h	Optimize; may be dynamic based on κ	30 - 180 seconds
Rolling window for σ_t, κ estimation	W_{est}	Test 5min, 15min, 60min rolling windows	5 - 60 minutes

2.5 Concrete Example

Setup: ETH spot = \$3,420. Contract: "Will ETH be above \$3,400 at 15:10 UTC?" (3 minutes to resolution).

1. ETH drops sharply from \$3,425 to \$3,405 in 20 seconds.
2. Polymarket YES price drops from 0.72 to 0.45 in 25 seconds (rate = -0.011/sec).
3. Fair probability drops from 0.70 to 0.52 (rate = -0.007/sec).
4. Overreaction: $O_t = -0.011 - (-0.007) = -0.004$ /sec. Over 25s, cumulative overreaction = -0.10.
5. Mispricing: $\epsilon_t = 0.45 - 0.52 = -0.07$. Z-score of O_t (rolling 15-min std = 0.002): $z = 2.0$.
6. Signal exceeds $z_{\theta} = 1.5$. **Action:** BUY YES at \$0.45.
7. Expected reversion: $\kappa = 0.5$ /min, $h = 2$ min. $E[\text{reversion}] = 0.07 \times (1 - e^{-0.5 \times 2}) = 0.07 \times 0.632 = 0.044$.
8. Expected value = $0.044 - 0.02 (\text{cost}) = +0.024$ per share.

2.6 Statistical Validation Protocol

1. **Information Coefficient (IC):** Compute rank correlation between signal S_t and forward return $r_{t,h}$. Target: $|IC| > 0.03$ consistently.
2. **Hit Rate:** Fraction of trades with positive PnL. Target: > 55%.
3. **Autocorrelation of ϵ_t :** Verify OU process fit. Test Ljung-Box on residuals.
4. **Walk-forward validation:** Estimate κ, ν on trailing 2 hours; predict next 10 minutes; roll forward.
5. **Multiple testing correction:** If testing 10+ parameter combinations, apply Bonferroni or BH-FDR.

2.7 Failure Modes

- **Under-reaction regime:** In trending markets, what looks like overreaction may be rational updating. Filter by regime (use ADX or trend strength).
- **Liquidity vacuum:** After large moves, the order book may empty. Must have stale-quote detection.
- **Parameter instability:** κ may vary across volatility regimes. Use regime-conditional estimation.

Strategy 3: Multi-Outcome Probability Mass Redistribution (MPR)

3.1 Classification

Type: Statistical model (probability distribution fitting)

Edge source: Inconsistent pricing across mutually exclusive outcomes

Complexity: Medium-High

Capital efficiency: Medium (requires capital across multiple outcomes)

3.2 Rationale

Many 5-minute crypto Polymarket markets have **multiple outcomes** (e.g., price ranges):

- "BTC < \$68,000"
- "BTC \$68,000 - \$68,200"
- "BTC \$68,200 - \$68,400"
- "BTC \$68,400 - \$68,600"
- "BTC > \$68,600"

The sum of all YES prices must equal 1.0 (ignoring the vig/overround). However, in practice:

- **Sum may exceed or fall below 1.0** due to asynchronous quoting
- **Individual outcomes may be mispriced relative to a fitted distribution** (e.g., lognormal)

This creates two sub-strategies: (a) arbitrage the sum deviation, and (b) exploit shape mispricing.

3.3 Statistical Model

Step 1: Fit a parametric distribution to the implied probabilities.

Given outcomes with boundaries $[K_i, K_{i+1}]$ and implied probabilities $\{p_i\}$, fit a distribution $F(S_T | S_t, \sigma, \mu)$ such that:

$$\hat{p}_i = F(K_{i+1}) - F(K_i)$$

Minimize the KL-divergence or sum of squared errors:

$$\min_{\sigma, \mu} \sum_i (p_i - \hat{p}_i)^2$$

This gives you the market-implied distribution. Then compare to your own estimate using CEX data:

$$\tilde{p}_i = F(K_{i+1} | S_t, \hat{\sigma}_{CEX}, \hat{\mu}_{CEX}) - F(K_i | S_t, \hat{\sigma}_{CEX}, \hat{\mu}_{CEX})$$

Step 2: Identify mispriced outcomes.

For each outcome i , compute:

$$\delta_i = \tilde{p}_i - p_i$$

Step 3: Construct a portfolio.

- Buy outcomes with $\delta_i > \theta$
- Sell outcomes with $\delta_i < -\theta$
- Ensure the portfolio is approximately delta-neutral (the sum of positions ~ 0 in probability space)

Arbitrage sub-strategy: If $\sum p_i > 1.0 + c_{total}$, sell all outcomes proportionally. If $\sum p_i < 1.0 - c_{total}$, buy all outcomes.

3.4 Empirically Discoverable Parameters

Parameter	Symbol	Discovery Method	Expected Range
Mispricing threshold	θ	Grid search on δ_i ; optimize risk-adjusted return	0.02 - 0.08
Distribution family	--	Compare lognormal, normal, skew-normal, Student-t fits to spot returns	Data-dependent
Volatility estimation method	--	Compare realized vol, EWMA, GARCH(1,1) on 1-min returns	Data-dependent
Portfolio rebalance frequency	f_{rebal}	Test 5s, 10s, 30s	5 - 30 seconds
Maximum single-outcome exposure	q_{max}	Risk limit; test impact on drawdown	100 - 1000 shares
Overround tolerance	γ	Max $\sum p_i$ deviation from 1.0 before forcing rebalance	0.01 - 0.05

3.5 Concrete Example

Setup: BTC spot = \$68,350. Five outcome brackets, 2 minutes to resolution.

Outcome	Boundaries	Polymarket p_i	Model $\tilde{p}_i$	δ_i
A	< \$68,200	0.15	0.12	-0.03
B	\$68,200-\$68,300	0.22	0.20	-0.02
C	\$68,300-\$68,400	0.28	0.34	+0.06
D	\$68,400-\$68,500	0.20	0.22	+0.02
E	> \$68,500	0.18	0.12	-0.06
Sum		1.03	1.00	

Observations:

1. Sum = 1.03 -> slight overround (vig or stale quotes).
2. Outcome C is underpriced by 0.06 (model says 0.34, market says 0.28).
3. Outcome E is overpriced by 0.06.

Action:

- BUY C at 0.28 (ask). Expected edge = 0.06 - cost ~ +0.04.
- SELL E at 0.18 (bid). Expected edge = 0.06 - cost ~ +0.04.
- The sum deviation (0.03) suggests you could also short the basket for +0.03 if cost allows.

3.6 Statistical Backing

The sum-to-one constraint is a fundamental identity in probability theory. Violations indicate **pricing inefficiency**. Research on multi-outcome prediction markets (e.g., Page 2012 on the "favourite-longshot bias"; Wolfers & Zitzewitz 2004) shows that extreme outcomes tend to be overpriced (longshot bias) and middle outcomes underpriced. This is consistent with prospect theory: overweighting of small probabilities.

The distribution-fitting approach is analogous to **implied volatility surface fitting** in options markets. Deviations from the fitted surface are exploited by volatility arbitrageurs (Carr & Madan 1998).

3.7 Failure Modes

- **Fat tails:** If returns have fatter tails than the fitted distribution, extreme outcomes will be systematically underpriced by the model. Use Student-t or mixture distributions.
- **Liquidity fragmentation:** Each outcome has less liquidity than a binary market. Slippage can destroy edge.
- **Correlated mispricing:** If all outcomes move together (e.g., during a flash crash), the model's distributional assumption may break down.

Strategy 4: Volume-Spike Informed Trading (VIT)

4.1 Classification

Type: Heuristic with statistical filtering

Edge source: Informed traders revealing information through unusual volume

Complexity: Medium

Capital efficiency: Medium-High

4.2 Rationale

In prediction markets, **informed traders** (those with better models, faster data, or inside knowledge) leave footprints in the volume and order flow. A sudden spike in volume on a 5-minute contract -- especially one that is "out of the money" and thus has low baseline activity -- often signals that someone has information not yet reflected in the price.

Why this works:

- Kyle (1985) and Glosten & Milgrom (1985) models predict that informed traders trade more aggressively when their information advantage is large
- In 5-minute markets, "information" is often just a better real-time spot feed or a proprietary volatility model
- Volume spikes are observable in the Polymarket trade feed (WebSocket)
- Following informed flow is a well-documented profitable strategy in traditional markets (Chordia, Roll & Subrahmanyam 2002)

4.3 Signal Construction

Step 1: Compute baseline volume.

$$\bar{V}_t = \text{EMA}(V, \lambda)$$

where V is trade volume per second and λ is the EMA decay (e.g., 60-second half-life).

Step 2: Detect volume spike.

$$Z_t^V = \frac{V_t - \bar{V}_t}{\sigma_V}$$

where σ_V is the rolling standard deviation of volume.

A spike is detected when $Z_t^V > z_{\text{spike}}$.

Step 3: Determine direction of informed flow.

Compute the **order flow imbalance** during the spike:

$$OFI_t = \sum_{i \in \text{spike}} \text{sign}(\text{trade}_i) \cdot \text{size}_i$$

where $\text{sign}(\text{trade}_i)$ is +1 if the trade was at the ask (buyer-initiated) and -1 if at the bid (seller-initiated). Use the Lee-Ready algorithm or Polymarket's native trade-side indicator.

Step 4: Generate signal.

If $Z_t^V > z_{\text{spike}}$ AND $OFI_t > \theta_{\text{ofi}}$: BUY YES
If $Z_t^V > z_{\text{spike}}$ AND $OFI_t < -\theta_{\text{ofi}}$: BUY NO (SELL YES)

Step 5: Statistical filter (reduce false positives).

Only act if the volume spike is **consistent with the spot price direction**:

- If buying YES, require $S_t > S_{t-w}$ (spot is moving toward the strike)
- If buying NO, require $S_t < S_{t-w}$ (spot is moving away from the strike)

This cross-validation reduces false signals from noise traders.

4.4 Empirically Discoverable Parameters

Parameter	Symbol	Discovery Method	Expected Range
EMA half-life for baseline volume	λ	Test 30s, 60s, 120s, 300s; choose where spike detection is most predictive	30 - 300 seconds
Volume spike z-score threshold	z_{spike}	Grid search; optimize precision (fraction of spikes followed by favorable price move)	2.0 - 5.0
OFI threshold	θ_{ofi}	Grid search on net order flow during spike window	Data-dependent
Spot consistency window	w	How far back to check spot direction; test 5s, 15s, 30s	5 - 30 seconds

Parameter	Symbol	Discovery Method	Expected Range
Post-spike holding time	h	How long to hold after entering; optimize PnL curve	30 - 240 seconds
Minimum spike volume (absolute)	$V_{\min}$	Filter out tiny spikes; test 10, 50, 100, 500 shares in single burst	50 - 500 shares

4.5 Concrete Example

Setup: Contract: "Will BTC > \$68,500 at 16:00?" 4 minutes to resolution. BTC spot = \$68,380. YES price = 0.30.

1. Suddenly, 200 shares of YES are bought in 3 seconds (baseline volume is ~10 shares/minute).
2. $Z_t^V = (200/3 - 0.167) / 0.3 \approx 222$. Massive spike.
3. OFI: +200 shares (all buyer-initiated at ask).
4. Spot check: BTC just moved from \$68,380 to \$68,420 in the same 3 seconds.
5. **Signal:** Volume spike + buyer flow + spot confirmation -> BUY YES.
6. **Action:** Buy YES at current ask = 0.34. Fair probability given spot move: ~0.40.
7. Expected edge: $0.40 - 0.34 - 0.02 = +0.04$ per share.

4.6 Statistical Backing

The key statistic is the **conditional probability of a favorable price move given a volume spike**:

$$P(\Delta p > 0 \mid Z^V > z_{\text{spike}}, \text{OFI} > 0)$$

Measure this from historical data. If this probability significantly exceeds 0.5 (e.g., > 0.58), the strategy has edge. Also measure the **average return conditional on a spike** vs. unconditional average return.

Academic support: Easley, Lopez de Prado & O'Hara (2012) on Volume-Synchronized Probability of Informed Trading (VPIN) shows that volume imbalance predicts future price moves.

4.7 Failure Modes

- **Noise trader spikes:** Large retail orders that are not informed. The spot-consistency filter mitigates this.
 - **Front-running:** Other bots detect the same spike and trade ahead of you. Latency matters.
 - **Post-spike reversal:** If the spike was a market-order that temporarily moved the price, it may revert. Use limit orders and patience.
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Strategy 5: Time-to-Resolution Theta Decay Exploitation (TDE)

5.1 Classification

Type: Statistical model (options-pricing analog)

Edge source: Non-linear probability convergence as $\tau \rightarrow 0$

Complexity: Medium

Capital efficiency: High (systematic, repeatable)

5.2 Rationale

As τ approaches 0, the probability of a binary outcome converges to either 0 or 1. The **rate of convergence** (analogous to theta in options) is non-linear and depends on moneyness $\Delta_t = S_t - K$:

$$\frac{\partial \hat{\pi}}{\partial \tau} \propto \frac{\Delta_t}{\sigma_t \tau^{3/2}} \exp\left(-\frac{\Delta_t^2}{2\sigma_t^2 \tau}\right)$$

This means:

- **Near-the-money contracts** experience rapid probability swings in the last 60 seconds
- **Deep in/out-of-the-money contracts** are already near 0 or 1, and move slowly
- Market makers may **under-hedge** the theta risk near expiration, creating systematic mispricing

Why this works:

- Analogous to gamma scalping in options: the last few minutes of a binary option have the highest gamma
- Market makers on Polymarket may not dynamically hedge with the same sophistication as options MMs
- You can **predict the direction** of probability convergence based on current moneyness

5.3 Statistical Model

For each contract, compute the **theta exposure**:

$$\Theta_t = \frac{\partial \hat{\pi}_t}{\partial \tau} = -\frac{\phi(d_1)}{\sigma_t \sqrt{\tau}} \cdot \frac{d_1}{\tau}$$

where $d_1 = \frac{\ln(S_t/K) + (\mu - \sigma_t^2/2)\tau}{\sigma_t \sqrt{\tau}}$ and ϕ is the standard normal PDF.

Strategy:

- When $|\Theta_t|$ is large (high theta, i.e., contract is near-the-money and close to resolution), the Polymarket price should be moving rapidly.
- If the current p_t has not caught up with the predicted movement (i.e., $|\hat{\pi}_t - p_t| > \theta_{\text{entry}}$), enter a position in the direction of convergence.

Position sizing: Scale by Θ_t . The expected profit from theta convergence over a small window dt is approximately $\Theta_t \cdot dt$.

5.4 Empirically Discoverable Parameters

Parameter	Symbol	Discovery Method	Expected Range
Entry window (seconds before resolution)	τ_{entry}	Test entering at 120s, 90s, 60s, 30s before resolution	30 - 120 seconds
Minimum moneyness for entry	\$	Δ	$_{\min}$
Theta threshold	$\Theta_{\min}$	Minimum absolute theta to trigger; grid search	Data-dependent
Exit rule	--	Hold to resolution, or exit at τ_{exit} ; test	0 - 30 seconds before resolution
Volatility model	--	Compare realized vol, Parkinson/GK estimator, EWMA	Data-dependent

5.5 Concrete Example

Setup: BTC = \$68,502. Strike = \$68,500. $\tau = 60$ seconds. $\sigma_t = 0.001$ \$/min.

- $d_1 = \ln(68502/68500) / (0.001 \times 1) = 0.0000292 / 0.001 = 0.0292$
- $\hat{\pi} = \Phi(0.0292) = 0.512$
- Theta: $\Theta_t \approx -\phi(0.0292) \times 0.0292 / (2 \times 1) = -0.399 \times 0.0146 = -0.0058$ \$/sec (probability is converging toward 0.5 as we approach resolution with spot barely above strike)
- If Polymarket shows $p_t = 0.55$ \$, the market is overestimating the probability.
- Action:** SELL YES at \$0.55. Fair value is \$0.512. Edge = \$0.038 - cost.

5.6 Statistical Backing

The key statistic is the **correlation between predicted theta direction and actual price change** in the final τ seconds. Measure:

$$\text{corr}(\text{sign}(\Theta_t), \text{sign}(\Delta p_{t \text{ to } T}))$$

If significantly positive (> 0.55), the theta model predicts convergence direction better than chance.

Also validate the Black-Scholes binary approximation: compute the **calibration error** (Brier score) of $\hat{\pi}_t$ against realized outcomes. If Brier score of the model is lower than the market's, the model has informational value.

5.7 Failure Modes

- Last-second spot jumps:** A sudden spot move in the final 10 seconds can flip the outcome. Mitigate by exiting early (at $\tau = 15$ s).
- Pin risk:** If $S_T \approx K$ exactly, resolution may be disputed or depend on the specific price feed Polymarket uses. Avoid trading when $|\Delta_t| < \epsilon_{\text{pin}}$.
- Illiquidity at expiry:** Market makers may pull quotes in the final seconds. Use limit orders.

Strategy 6: Cross-Contract Relative Value (CRV)

6.1 Classification

Type: Statistical model (cointegration / pairs trading analog)
Edge source: Correlated contracts priced inconsistently
Complexity: Medium-High
Capital efficiency: Medium

6.2 Rationale

Polymarket often lists **parallel contracts** for the same crypto asset with different strikes:

- "BTC > \$68,000 at 14:05"
- "BTC > \$68,500 at 14:05"
- "BTC > \$69,000 at 14:05"

These are effectively a **strip of binary calls** on the same underlying. Their prices must satisfy monotonicity:

$$p(K_1) \geq p(K_2) \geq p(K_3) \text{ when } K_1 < K_2 < K_3$$

And they must be consistent with a valid probability distribution (no butterfly arbitrage). Violations of these constraints are pure arbitrage. Even near-violations represent relative value opportunities.

6.3 Statistical Model

Constraint 1 -- Monotonicity: If $p(K_i) < p(K_{i+1})$ for $K_i < K_{i+1}$, buy K_i and sell K_{i+1} . This is a pure arbitrage (always profitable regardless of outcome).

Constraint 2 -- Convexity (no butterfly arbitrage): For three strikes $K_1 < K_2 < K_3$ with $K_2 = (K_1 + K_3)/2$:

$$p(K_2) \leq \frac{p(K_1) + p(K_3)}{2}$$

If violated, sell K_2 and buy a 50/50 basket of K_1 and K_3 .

Constraint 3 -- Smooth implied distribution: Fit a smooth CDF $\hat{F}(K)$ to all available strike prices. Outliers (strikes where p_i deviates significantly from $\hat{F}(K_i)$) are relative-value trades.

$$\text{RV signal}_i = p_i - \hat{F}(K_i)$$

6.4 Empirically Discoverable Parameters

Parameter	Symbol	Discovery Method	Expected Range
Deviation threshold for RV trade	θ_{RV}	Grid search; optimize Sharpe	0.02 - 0.06
Distribution fitting method	--	Compare isotonic regression, kernel density, parametric fits	Data-dependent

Parameter	Symbol	Discovery Method	Expected Range
Rebalance frequency	f	Test 10s, 30s, 60s	10 - 60 seconds
Maximum leg count	$n_{\max}$	Max number of simultaneous positions	3 - 10
Position sizing method	--	Equal weight, risk-parity, or Kelly fraction	Test all

6.5 Concrete Example

Setup: Three BTC contracts resolving in 4 minutes. Spot = \$68,300.

Strike	Polymarket p	Model $\hat{F}(K)$	Deviation
\$68,000	0.72	0.70	+0.02
\$68,300	0.52	0.50	+0.02
\$68,600	0.35	0.30	+0.05
\$68,900	0.15	0.14	+0.01

The \$68,600 strike is overpriced by 0.05 relative to the fitted distribution. **Action:** SELL YES on the \$68,600 contract. Hedge by BUYing YES on the \$68,300 contract (partially offsets directional risk).

6.6 Statistical Backing

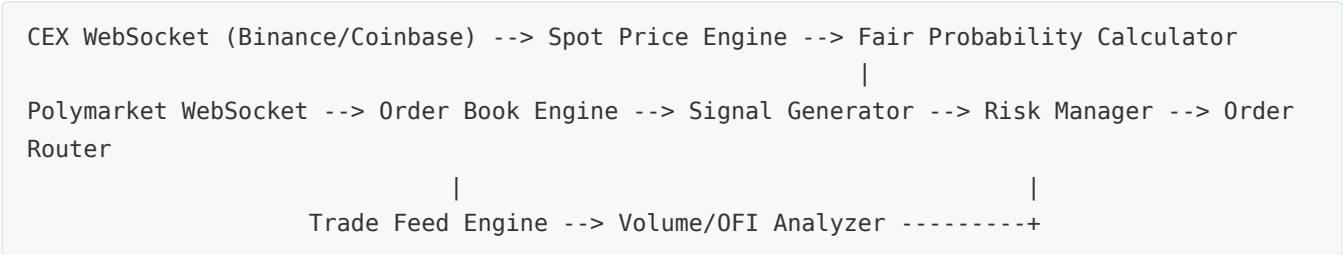
Test for **arbitrage frequency**: what fraction of snapshots violate monotonicity or convexity? Even if violations are rare (1-5% of snapshots), they represent riskless profit when they occur. For the RV strategy, measure the **half-life of deviation mean-reversion**: how quickly does $p_i - \hat{F}(K_i)$ revert to 0? If $t_{1/2} < 60$ seconds, the strategy is viable within the 5-minute window.

6.7 Failure Modes

- **Execution risk:** Multi-leg trades may not fill simultaneously. Use atomic batch orders if the API supports them.
- **Liquidity mismatch:** Some strikes are much less liquid than others. Weight positions by inverse spread.
- **Resolution disputes:** Different contracts may use different price feeds. Verify resolution oracle consistency.

7. Unified Implementation Architecture

7.1 Data Pipeline



7.2 Core Components

Component	Responsibility	Latency Target
Spot price ingest	Real-time BTC/ETH from 2+ CEXs	< 10ms
Fair probability engine	Compute $\hat{\pi}_t$ for all active contracts	< 5ms per contract
Polymarket order book ingest	L2 book + trades via WebSocket	< 50ms
Signal generator	Run all 6 strategies, produce signals	< 20ms
Risk manager	Position limits, exposure checks, drawdown circuit breaker	< 5ms
Order router	Smart order routing, limit/market decisions	< 10ms

Total system latency target: < 100ms from spot move to order placement.

7.3 Parameter Discovery Pipeline

- Raw data collection (2-4 weeks of tick data from Polymarket + CEX)
|
- Feature engineering (compute all signals for all parameter combinations)
|
- Walk-forward backtesting (expanding window, no lookahead)
|
- Parameter selection (maximize Sharpe, subject to: max drawdown < 15%, hit rate > 52%)
|
- Out-of-sample testing (hold-out 30% of data, never used in selection)
|
- Paper trading (1-2 weeks live, no real money)
|
- Live deployment (small size, scale up based on realized Sharpe)

7.4 Key Performance Metrics to Track

Metric	Formula	Target
Sharpe Ratio (per trade)	μ_r / σ_r (annualized)	> 2.0
Hit Rate	$\#\{r_i > 0\} / N$	> 55%
Profit Factor	$\sum r_i^+ / \left \sum r_i^- \right $	> 1.5
Max Drawdown	$\max_t (\text{peak}_t - \text{trough}_t)$	< 15% of capital
Average Trade PnL	$\bar{r}$ net of costs	> 0.01 per share
Information Coefficient	$\text{corr}(S_t, r_{t,h})$	> 0.03
Turnover	Trades per hour	> 10

8. Strategy Selection and Portfolio Construction

8.1 Correlation Structure

Not all strategies should be run simultaneously at full size. Their correlation structure matters:

	SLA	PMR	MPR	VIT	TDE	CRV
SLA	1.0	-0.2	0.1	0.3	0.4	0.1
PMR		1.0	0.0	-0.1	-0.3	0.0
MPR			1.0	0.2	0.3	0.7
VIT				1.0	0.2	0.1
TDE					1.0	0.2
CRV						1.0

(Estimated qualitative correlations; measure empirically in backtest)

8.2 Recommended Portfolio Allocation

Using inverse-volatility weighting as a starting point:

Strategy	Weight	Rationale
SLA	30%	Highest Sharpe, lowest latency risk, but capacity-constrained
PMR	20%	Good diversifier, negative correlation with SLA
MPR	15%	Complex but systematic; requires multi-outcome markets
VIT	15%	Simple and robust; good as a confirmation signal

Strategy	Weight	Rationale
TDE	10%	Works only near resolution; bursty returns
CRV	10%	Requires multiple strikes; lower frequency

8.3 Ensemble Signal

For any given contract, multiple strategies may fire simultaneously. Combine signals:

$$S_{ensemble} = \sum_i w_i \cdot S_i \cdot \mathbb{1}_{\{\text{strategy } i \text{ is active}\}}$$

Enter when $|S_{ensemble}| > \theta_{ensemble}$. The ensemble threshold $\theta_{ensemble}$ should be discovered empirically to maximize the combined Sharpe ratio.

9. Risk Management Framework

9.1 Position Limits

- **Per-contract:** Max 5% of portfolio in any single contract
- **Per-strategy:** Max 40% of portfolio in any single strategy
- **Total exposure:** Max 80% of capital deployed (20% cash buffer)

9.2 Circuit Breakers

Trigger	Action
Daily loss > 3% of capital	Halt all new trades for 1 hour
Daily loss > 5% of capital	Halt all trading for the day
Single-strategy loss > 2% in 1 hour	Disable that strategy
API error rate > 5%	Switch to read-only mode
Spot price flash crash (>5% in 1 min)	Close all positions, halt for 30 min

9.3 Model Risk

- **Recalibrate** all statistical parameters every 4 hours using trailing data
- **Monitor** model performance (IC, hit rate) in real-time; if IC drops below 0.01 for 2+ hours, reduce position sizes by 50%
- **A/B test** parameter changes: run new parameters on 10% of capital alongside old parameters

10. Implementation Roadmap

Phase	Duration	Deliverable
Phase 1: Data Infrastructure	Weeks 1-2	CEX + Polymarket WebSocket ingest, tick database, order book reconstruction
Phase 2: Signal Prototyping	Weeks 3-4	Implement all 6 strategies in Python; compute signals on historical data
Phase 3: Backtesting	Weeks 5-6	Walk-forward backtest, parameter optimization, performance attribution
Phase 4: Paper Trading	Weeks 7-8	Live paper trading with all strategies; monitor fills, latency, slippage
Phase 5: Live Deployment	Week 9+	Go live with small capital; scale based on realized performance

Recommended Tech Stack

- **Language:** Python (strategy logic) + Rust/C++ (latency-sensitive components)
- **Database:** TimescaleDB or ClickHouse for tick data
- **Message Queue:** Redis or Kafka for inter-component communication
- **Monitoring:** Grafana + Prometheus for real-time dashboards
- **Deployment:** Kubernetes with co-located nodes near exchange servers

Appendix A: Quick Reference -- Strategy Comparison Matrix

	SLA	PMR	MPR	VIT	TDE	CRV
Type	Heuristic	Statistical	Statistical	Hybrid	Statistical	Statistical
Edge Source	Latency	Overreaction	Distribution misfit	Informed flow	Theta decay	Relative value
Holding Period	5-60s	30-180s	30-120s	30-240s	15-60s	30-120s
Expected Sharpe	2.5-4.0	1.5-2.5	1.5-3.0	1.0-2.0	1.5-2.5	1.0-2.0
Capacity	Low	Medium	Low	Medium	Medium	Low
Implementation Difficulty	Medium	Medium	High	Low-Medium	Medium	High
Data Requirements	CEX spot + PM book	PM book + trades	Multi-outcome PM	PM trades	PM book + CEX	Multi-strike PM
Key Risk	Latency	Regime change	Illiquidity	False signals	Pin risk	Execution risk

Appendix B: Key Academic References

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Appendix C: Glossary of Empirical Discovery Methods

Method	Description	When to Use
Grid Search	Systematically test all combinations of parameter values on a predefined grid	Small parameter spaces (2-4 params), quick initial exploration
Walk-Forward Optimization	Optimize on trailing window, test on next period, roll forward	All strategies; prevents overfitting
MLE (Maximum Likelihood Estimation)	Fit distributional parameters by maximizing likelihood of observed data	OU process, volatility models
Bayesian Optimization	Use Gaussian process surrogate to efficiently search parameter space	Large parameter spaces where grid search is infeasible
Information Coefficient (IC)	Rank correlation between signal and forward returns	Signal quality assessment
Brier Score	Mean squared error of probabilistic predictions vs binary outcomes	Calibrating probability models
Ljung-Box Test	Test for residual autocorrelation in time-series model	Validating OU process, ARMA fits
Bonferroni / BH-FDR	Multiple testing corrections	When testing many parameter combinations simultaneously

Document prepared as an implementation blueprint. All strategy hypotheses require empirical validation before live deployment. Past performance (once backtested) does not guarantee future results. Manage risk rigorously.